

Lab Report: Week 3 - Simple Pendulum

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2026-08-12

Imports and Functions

All calculations were done in Python. Starting with imports:

```
import matplotlib.pyplot as plt
from uncertainties import ufloat
import numpy as np
from scipy import odr
import pandas as pd
```

Defining the gravity constant in Canberra:

```
g = 9.796
```

To handle our data properly, we used a custom Python function (`odr_fit`) leveraging `scipy.odr`. Because our measurements have uncertainties in both the x and y axes, standard linear regression won't cut it. This function extracts the nominal values and errors from our `uncertainties` objects, sets up a linear model ($y = mx + c$), and performs an ODR to find the best-fit slope, its uncertainty, and the y-intercept.

```
def odr_fit(x_data, y_data, zero_intercept: bool = False):
    x = np.array([v.n for v in x_data])
    sx = np.array([v.s for v in x_data])
    y = np.array([v.n for v in y_data])
    sy = np.array([v.s for v in y_data])

    data = odr.RealData(x, y, sx=sx, sy=sy)

    if zero_intercept:
        linear_model = odr.Model(lambda p, x: p[0] * x)
        beta0 = [10.0]
    else:
```

```

linear_model = odr.Model(lambda p, x: p[0] * x + p[1])
beta0 = [10.0, 0.0]

odr_result = odr.ODR(data, linear_model, beta0=beta0).run()

slope = odr_result.beta[0]
slope_err = odr_result.sd_beta[0]

if zero_intercept:
    intercept = 0.0
    intercept_err = 0.0
else:
    intercept = odr_result.beta[1]
    intercept_err = odr_result.sd_beta[1]

return slope, slope_err, intercept, intercept_err, (x, y, sx, sy)

```

To visualize our results, we wrote a plotting function (`plot`) using `matplotlib`. This function creates a clear scatter plot of our data complete with error bars on both axes, overlays the best-fit line calculated from our ODR regression, and formats the legend to display our final spring constant value and its uncertainty.

```

def plot(x, y, sx, sy, slope, slope_err, intercept, xlabel, ylabel, title,
filename, zero_intercept=False):
    x_fit = np.linspace(min(x), max(x), 100)
    y_fit = slope * x_fit + intercept

    plt.figure(figsize=(7, 5))
    plt.errorbar(
        x, y,
        xerr=sx, yerr=sy,
        fmt='o',
        color="navy",
        ecolor="crimson",
        capsize=3,
        label="Experimental Data"
    )
    plt.plot(
        x_fit, y_fit,
        "k--",
        label=f"Fit: Slope = {slope:.3f} ± {slope_err:.3f}"
    )
    plt.xlabel(xlabel)

```

```

plt.ylabel(ylabel)
plt.title(title)
plt.grid(True, alpha=0.6)
plt.legend()
plt.tight_layout()
plt.savefig(filename, dpi=300)
plt.show()

```

To check if both methods gave consistent results, we run a sigma test (`sigma_test`). It calculates the statistical difference between the two spring constant values relative to their combined uncertainties. If the difference is less than or equal to 3 standard deviations (3σ), it indicates our results agree within experimental error.

```

def sigma_test(k1, sk1, k2, sk2):
    sigma_diff = abs(k1 - k2) / np.sqrt(sk1**2 + sk2**2)
    print("SIGMA TEST")
    print(f"Static k   : {k1:.3f} +/- {sk1:.3f} N/m")
    print(f"Dynamic k  : {k2:.3f} +/- {sk2:.3f} N/m")
    print(f"Discrepancy: {sigma_diff:.2f} sigma")

    if sigma_diff <= 3.0:
        print("results AGREE within experimental uncertainty")
    else:
        print("results DISAGREE")

```

Pre-lab Questions

Question 1-

Using equation (3),

$$T = 2\pi\sqrt{\frac{L}{g}}$$

$$\frac{T}{2\pi} = \sqrt{\frac{L}{g}}$$

$$\left(\frac{T}{2\pi}\right)^2 = \frac{L}{g}$$

$$g = \frac{4(\pi)^2 L}{T^2}$$

Question 2-

Using the above obtained equation for g , and substituting measured values with their absolute errors, $L \pm \sigma_L$ and $T \pm \sigma_T$,

$$g \pm \sigma_g = \frac{4(\pi)^2(L \pm \sigma_L)}{(T \pm \sigma_T)}$$

Rearranging,

$$\sigma_g = \pm \left(\frac{4(\pi)^2(L \pm \sigma_L)}{(T \pm \sigma_T)} - g \right)$$

Experiment 1

The aim of this experiment is to measure the period of a simple pendulum with respect to some initial angle to highlight the connection between angle, oscillatory period, and the local acceleration due to gravity.

Experiment 2